functional-design | Status: current | App v1.0.20 | Reviewed 2026-10-24

Functional Design

1. Product scope

The app is a showcase and sandbox for **on-device generative AI**. Users browse tasks (features), download a compatible model per task, and run it offline. Inference never leaves the device. Network access is used only for model and catalog downloads, optional telemetry, user-added remote skills and MCP servers, and push notifications.

Supported OS: Android 12+ (`minSdk 31`). An iOS app and a macOS build exist (see the root README), but their source is **not** in this repository.

2. Actors

Actor	Description
End user	Runs tasks, manages models, adds skills and MCP servers
Skill author	Writes `SKILL.md` and JS skills and shares them through GitHub Discussions or a URL
Catalog maintainer	Edits `model_allowlists/<version>.json` to add, enable, or disable models
Developer	Builds the app, adds custom tasks or functions

3. Feature catalog

Model lists come from `model_allowlists/1_0_20.json` (`taskTypes`).

Feature (task id)	User value	Models (1.0.20 allowlist)	Inputs	Key limits
AI Chat (`llm_chat`)	Multi-turn chat with optional Thinking Mode	Gemma-4-E2B/E4B-it, Gemma-3n-E2B/E4B-it, Gemma3-1B-IT, Qwen2.5-1.5B-Instruct, DeepSeek-R1-Distill-Qwen-1.5B	Text, image and audio when the model supports them, voice dictation	Context = model `maxTokens`

Feature (task id)	User value	Models (1.0.20 allowlist)	Inputs	Key limits
Ask Image (`llm_ask_image`)	Questions about photos or camera captures	Gemma-4 and Gemma-3n families	Text + images	≤10 images per session (`MAX_IMAGE_COUNT`); 1 for AICore
Audio Scribe (`llm_ask_audio`)	Transcribe or translate speech	Gemma-4 and Gemma-3n families	Recorded or picked audio	1 clip (`MAX_AUDIO_CLIP_COUNT`), ≤30 s, 16 kHz
Prompt Lab (`llm_prompt_lab`)	Single-turn prompting with templates and sampler control	All chat LLMs	Text	Templates: Free form, Rewrite tone, Summarize text, code snippet languages (C++, Java, JavaScript, Kotlin, Python, ...)
Agent Skills (`llm_agent_chat`)	Chat that can use skills, MCP tools, and device intents	Gemma-4-E2B/E4B-it	Text, image, audio	≤15 recommended selected skills (`MAX_RECOMMENDED_SKILL_COUNT`)
Instant Media Search (`smart_album`)	Natural-language and image-to-image search over the photo library	EmbeddingGemma-2	Search text, a reference photo	Needs indexing first; user library or built-in sample album
Video Moment Finder (`video_moment_finder`)	Find moments in videos by description, trim and export clips	EmbeddingGemma-2	Videos, search text	Per-project processing config (segment length, frames per segment, process sound track)
Mobile Actions (`llm_mobile_actions`) — experimental	Offline device control by voice or text	MobileActions-270M (FunctionGemma)	Text	7 functions (see 02-agents §9)

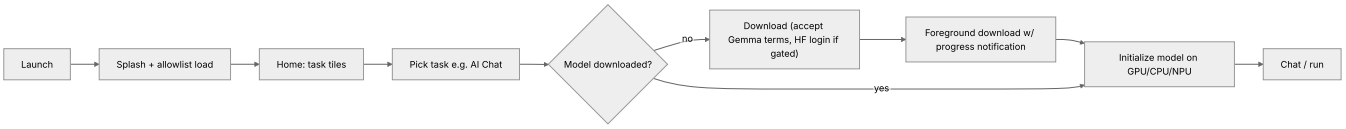
Feature (task id)	User value	Models (1.0.20 allowlist)	Inputs	Key limits
Tiny Garden (`llm_tiny_garden`) — experimental	Natural-language mini-game	TinyGarden-270M (FunctionGemma)	Text	Conversation resets after N turns (default 3, range 1–30)
Scrapbook (`mp_scrapbook`)	Cut objects out of photos and make collages	"Magic touch" (MediaPipe InteractiveSegmenter)	Photo + touch strokes	—
Test chat (`llm_test`)	Chat with a side-loaded or imported model	User-imported `.litertlm` / `.task`	Text, image, audio	—

Cross-cutting features:

Feature	Behaviour
Model manager	Lists allowlist models and imported models. Download, cancel, and delete. Toggle optional components (extra data files). Shows a memory warning when a model needs more RAM (`minDeviceMemoryInGb`) than the device has.
Model import	Import a local `.litertlm` (or `.task`) file with LLM configs (accelerators, max tokens, sampling defaults, image/audio/thinking support). Browse and search Hugging Face, or import by Hugging Face URL.
Hugging Face sign-in	For gated models, a probe returns `GATED` or `NEEDS_TOKEN_EXCHANGE`. The app runs an OAuth (AppAuth) login with scope `read-repos` and stores the token.
Benchmark	LiteRT-LM models only. Configurable prefill/decode tokens and runs. Results are stored and can be compared.
Chat history	Sessions persist per task. A side sheet lets users resume or delete them.
Settings	Theme (light/dark/auto), Firebase Analytics opt-out, Terms of Service, Gemma Terms, open-source licenses
Terms gating	App ToS dialog. Gemma Terms of Use must be accepted before downloading Gemma models (`DownloadAndTryButton`).
Notifications screen	Lists local notifications scheduled by skills (`schedule_notification`) and allows deleting them
New release banner	Checks `api.github.com/repos/<repo>/releases/latest` and prompts when a newer version exists
Deep links	Open a task or model directly, optionally prefilled with a query that the user must confirm (see 01-architecture §7)

4. Primary user flows

4.1 First run → first answer



4.2 Agent Skills: add and use a skill

1. Open Agent Skills, then the Skill manager.
2. Add a skill from Featured (empty in this build), from a URL, or from a local folder. Accept the disclaimer.
3. Enable or disable skills. Edit or test a skill in the Skill tester.
4. Ask a question. The model loads the matching skill and may run JS (hidden WebView), an intent, or an MCP tool.
5. If the skill needs a secret, the user is asked once and the secret is stored on the device.
6. Results show as text, an image, or an embedded webview card.

4.3 Agent Skills: add an MCP server

1. MCP chip, then Manage MCP servers, then **+**, then Add from URL (accept the disclaimer).
2. Enter the server URL (StreamableHTTP, usually `.../mcp`) and optionally an authorization header (name and value).
3. The app connects and lists the server's tools. The user enables or disables each tool, or turns on *always allow*.
4. On each call to a tool that isn't *always allow*, the user chooses Deny, Allow once, or Always allow.

4.4 Instant Media Search

1. Onboarding dialog, then choose the sample album or the user's library (full or partial media access).
2. Indexing runs in a background `SmartAlbumIndexingWorker` with a progress card and notification.
3. Search-as-you-type returns photos ranked by similarity. "Search similar" does image-to-image search.

4.5 Video Moment Finder

1. Create a project and add videos.

2. Adjust processing (segment duration, frames per segment, process sound track), then process.
3. Search moments in natural language. Choose top-K. Results show on a timeline.
4. Open a moment, trim the clip, save it to "Saved moments", or export.

4.6 Scrapbook

1. Pick a photo. Draw positive or negative strokes, and the segmenter produces a cutout.
2. Style cutouts (border, fill, doodles), arrange them in a collage or grid, set a coloured background.
3. Save to the device album.

5. Error and edge behaviour (user-visible)

Situation	Behaviour
Allowlist fetch fails	Falls back to the cached allowlist. If there is none, sets the error "Failed to load model list", which the home screen shows.
Local test allowlist present	Shows a warning dialog that `/data/local/tmp/model_allowlist_test.json` is in use
Device not on the AICore device list	AICore models are hidden
NPU-only model on an unsupported SoC	Model hidden
Pixel devices	"NPU" is relabelled "TPU". GPU is removed on Pixel 10.
Download interrupted	Resumes from the partial `.gallerytmp` file on next launch (HTTP `Range`)
Model init fails	Error dialog with a cleaned-up runtime message
Not enough memory	Warning before download or entry (`MemoryWarning`)
Skill or tool not found	The model gets a hint to try it as a skill instead
MCP tool denied	Tool returns "Permission denied by user"

6. Out of scope (this repository)

- iOS and macOS app source.
- Server-side components. The only "backend" is static JSON on GitHub (`model_allowlists/`) and GitHub Pages for `skills/`.
- Cloud inference.

See also: 09-requirements, 02-agents.